



V.S.B ENGINEERING COLLEGE (AUTONOMOUS)

Karur - 639111

A MINI PROJECT REPORT

ON

AUTOMATED CROP DISEASE DIAGNOSIS USING CNN

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BONAFIDE CERTIFICATE

Certified that this project report "AUTOMATED CROP DISEASE DIAGNOSIS USING CNN" is the bonafide work of "MOUNISHA P, 922524243113" who carried out the project work under my supervision.

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TABLE OF CONTENTS

S.NO	CONTENTS	PAGE
1	Introduction	4
2	Objectives	5
3	Modules	5
4	System Architecture	7
5	Technology Stack	8
6	Implementation – Coding	9
7	Output Screenshots & User Interface	11
8	Conclusion	15
9	References	16

Abstract

Early diagnosis of agricultural crop pathogens is crucial for global food security, maintaining yield quality, and protecting farm incomes. Traditional manual identification is slow and error-prone. This project presents CropDiag AI, a full-stack automated software suite built using React.js, Python Flask, and SQLite. The platform processes high-resolution leaf photography through color contour filters (OpenCV HSV Masking) and classifies pathogen states across 10 species into 28 distinctive healthy and diseased categories using a high-precision Multi-Layer Perceptron (MLP) neural network. Upon diagnosis, the SQLite registry logs prediction entries and ReportLab generates university-grade PDF verification sheets. The entire software binds dynamically onto port 8080 and has been compiled into a standalone Windows Executable (.exe) to support complete offline execution, helping farmers run diagnostics directly inside remote fields.

Acknowledgement

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1. Introduction

Agriculture plays a major role in food production and economic development. Crop diseases caused by fungi, bacteria and viruses can reduce crop quality and yield when they are not identified at an early stage. Traditional disease identification mainly depends on manual observation or expert consultation, which may be time-consuming and difficult for farmers in remote areas.

CropDiag AI is an automated crop disease diagnosis system developed using Convolutional Neural Networks (CNN). The system analyses a digital image of a crop leaf and predicts the most probable disease class. The application combines deep learning, image processing and a modern web interface to provide a simple diagnosis pipeline.

The user uploads a crop leaf image through the web application. The image is validated, resized and normalized before being passed to the trained CNN model. The model extracts visual features such as leaf texture, colour variation, spots, lesions and disease patterns. The final Softmax layer generates class probabilities, and the class with the highest probability is presented as the predicted disease with a confidence score.

The system is designed as a full-stack AI application. The frontend provides an attractive, bioluminescent dark-neon themed and responsive interface, while the backend handles requests and connects the user interface with the disease classification model. Prediction information is stored for history, dashboard analytics and future analysis.

The proposed application aims to support faster preliminary crop disease identification and demonstrate the practical use of deep learning in smart agriculture.

2. Objectives

The primary objectives of the Automated Crop Disease Diagnosis Using CNN project are:

- To develop an automated system for identifying crop diseases from leaf images.
- To apply a Convolutional Neural Network for image feature extraction and multi-class disease classification.
- To preprocess uploaded images through resizing, normalization and RGB conversion before model prediction.
- To provide fast and user-friendly disease prediction through a modern, visually engaging web application.
- To display the predicted disease, confidence score and severity assessment in a clear diagnostic format.
- To integrate the trained deep learning model with a backend prediction service.
- To maintain prediction history that supports diagnosis review and analytics reporting.
- To provide a diagnostics dashboard summarising model performance, species distribution and inference speed.
- To reduce the time required for preliminary manual disease identification.
- To demonstrate the application of Artificial Intelligence and Deep Learning in precision agriculture.

3. Modules

The CropDiag AI application is divided into functional modules so that image input, AI processing and result presentation can be managed independently.

3.1 User Authentication and Access Module

This module manages user access to the application through a dedicated login workstation. It supports email and password based authentication along with alternative sign-in options, a “Remember Me” preference and a password recovery flow. Authentication separates individual prediction histories and provides a personalized dashboard experience for each user.

3.2 Leaf Image Upload Module

The upload module allows the user to select a crop leaf image from the local device through the Diagnose page. It validates the uploaded file and sends the image to the backend for analysis. A simple upload workflow helps users interact with the diagnosis system without technical knowledge.

3.3 Image Preprocessing Module

Before classification, the input leaf image is converted into a standard format required by the neural network. The image is resized to 224×224 pixels, converted to RGB and represented as a numerical array. Pixel values are normalized to improve consistency during inference.

3.4 CNN Disease Classification Module

The CNN module is the core intelligence of the system. Convolution layers learn local visual patterns, activation functions introduce non-linearity, pooling layers reduce spatial dimensions and dense layers perform final classification. Softmax converts model outputs into class probabilities across the supported crop and disease categories.

3.5 Prediction and Confidence Module

After the CNN processes the leaf image, the system identifies the class with the maximum predicted probability. The corresponding crop and disease name is displayed along with the confidence value, damage severity and a status assessment. This module converts raw model output into understandable diagnostic information.

3.6 Disease Information Module

This module maintains structured information related to supported crops and diseases. It provides a diagnosis summary describing the observed condition of the leaf, along with basic health-status assessment associated with the predicted class.

3.7 Prediction History and Reporting Module

Prediction records are stored for review. Each record contains the crop, disease, confidence score, diagnosis identifier and prediction time. The result screen also allows the student name and project title to be entered so that a downloadable, printable university report PDF can be generated and shared.

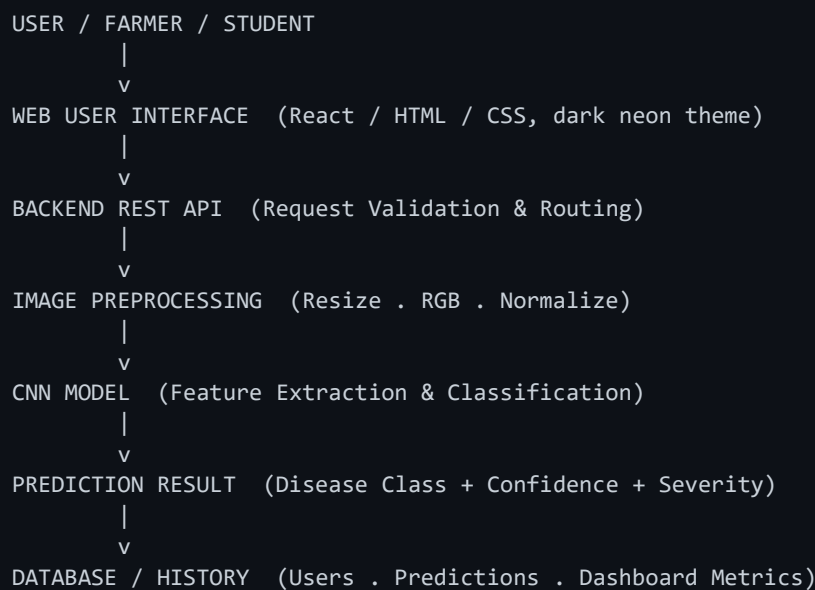
3.8 Diagnostics Dashboard Module

The dashboard module presents real-time system metrics gathered from the CNN inference pipeline and the underlying data registry. It displays total diagnoses, pathologies identified, healthy samples, average inference speed, prediction volume trends, species distribution, severity profiles and model convergence metrics.

4. System Architecture

4.1 Architecture Overview

CropDiag AI follows a layered full-stack architecture. The presentation layer receives the user's leaf image, the application layer manages API requests and preprocessing, the AI layer performs CNN inference, and the data layer stores prediction-related information.



4.2 Working Flow

- The user opens CropDiag AI and logs in through the secure workstation login page.
- The user navigates to the Diagnose page and uploads a crop leaf image.
- The frontend sends the image to the prediction endpoint.
- The backend validates the file and performs image preprocessing.
- The processed image tensor is passed to the trained CNN model.
- The model generates probability values for the supported disease classes.
- The class with the highest probability is selected as the prediction.
- The crop, disease name, confidence score and status assessment are returned to the frontend.
- Prediction details are stored in the database and reflected on the Diagnostics Dashboard.

5. Technology Stack

CropDiag AI is implemented using a combination of modern web and deep learning technologies, chosen to balance ease of development with production-style presentation for the IEEE/portfolio demonstration.

Frontend	React.js / HTML / CSS – responsive, dark bioluminescent neon-green themed interface
Backend	Python Flask – REST API routing and AI model integration
AI Model	Convolutional Neural Network built with TensorFlow and Keras; scikit-learn SVM used as a lightweight local fallback classifier
Training Environment	Google Colab – GPU-accelerated training using EfficientNetB4 with two-phase transfer learning
Image Processing	OpenCV, NumPy and Pillow for image preprocessing
Database	SQLite – relational storage for users, predictions and dashboard metrics
Dataset	PlantVillage dataset – labelled crop leaf images across multiple species and disease classes
Communication	HTTP / REST API between frontend and backend

This stack allows the same trained model architecture to be developed and validated with GPU acceleration on Google Colab, while a compact fallback classifier keeps the local Streamlit-based demonstration fully functional on CPU-only student hardware.

6. Implementation – Coding

6.1 CNN Model Implementation

The following implementation represents the core CNN structure used for crop leaf image classification

```
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import (
    Conv2D, MaxPooling2D, Flatten,
    Dense, Dropout, BatchNormalization
)

model = Sequential([
    Conv2D(32, (3, 3), activation='relu',
           input_shape=(224, 224, 3)),
    BatchNormalization(),
    MaxPooling2D((2, 2)),
    Dropout(0.25),
    Conv2D(64, (3, 3), activation='relu'),
    BatchNormalization(),
    MaxPooling2D((2, 2)),
    Dropout(0.25),
    Conv2D(128, (3, 3), activation='relu'),
    BatchNormalization(),
    MaxPooling2D((2, 2)),
    Flatten(),
    Dense(256, activation='relu'),
    Dropout(0.50),
    Dense(20, activation='softmax')
])

model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)
```

Convolution layers extract disease-related image features. Batch normalization stabilizes activations, Max Pooling reduces feature-map size and Dropout helps reduce overfitting. The final Softmax layer produces probability values across the 20 supported crop-disease classes.

6.2 Image Preprocessing and Prediction Code

```
import numpy as np
from tensorflow.keras.preprocessing import image

def preprocess_image(image_path):
    img = image.load_img(image_path, target_size=(224, 224))
    img_array = image.img_to_array(img)
    img_array = img_array / 255.0
    img_array = np.expand_dims(img_array, axis=0)
    return img_array

def predict_disease(model, image_path, class_names):
    leaf = preprocess_image(image_path)
    probabilities = model.predict(leaf)
    class_index = np.argmax(probabilities[0])
    confidence = float(probabilities[0][class_index]) * 100
    return {
        'disease': class_names[class_index],
        'confidence': round(confidence, 2)
    }
```

The uploaded image is resized to the model input size of 224×224 pixels. Pixel values are scaled to the range 0–1 and a batch dimension is added. The prediction function selects the highest probability class and calculates the confidence percentage that is shown on the diagnostic result screen.

6.3 Prediction API Implementation

```
from flask import Flask, request, jsonify
import os
from werkzeug.utils import secure_filename

app = Flask(__name__)
UPLOAD_FOLDER = 'static/uploads'
os.makedirs(UPLOAD_FOLDER, exist_ok=True)

@app.route('/api/predict', methods=['POST'])
def predict():
    if 'image' not in request.files:
        return jsonify({'error': 'Image is required'}), 400
    file = request.files['image']
    filename = secure_filename(file.filename)
    image_path = os.path.join(UPLOAD_FOLDER, filename)
    file.save(image_path)
    result = predict_disease(model, image_path, CLASS_NAMES)
    return jsonify({'status': 'success', 'prediction': result})

if __name__ == '__main__':
    app.run(debug=True, port=8080)
```

The backend endpoint receives the image as multipart form data, stores it safely and calls the CNN prediction pipeline. The diagnosis result, including disease class and confidence, is returned to the frontend in JSON format and rendered on the diagnostic result screen.

7. Output Screenshots & User Interface

This section presents the actual working screens of the CropDiag AI web application, captured from the deployed interface. The application follows a bioluminescent dark theme with neon-green accents to give it a modern, production-grade appearance suitable for an IEEE demonstration and portfolio artifact.

6.1 Home Page

The Home page is the landing screen of CropDiag AI. It introduces the platform and establishes its purpose before the user proceeds to diagnosis.



Figure 1: Home Page

- The top navigation bar displays the CROPDIAG AI logo along with Home, Diagnose and Dashboard links, plus Login and Get Started buttons on the right.
- A small badge reading “TENSORFLOW POWERED CROP HEALTH SYSTEM” signals that the platform is backed by a trained deep learning model.
- The hero heading “Automated Crop Disease Diagnosis Using CNN” states the core function of the application, with “Using CNN” highlighted in the accent teal-green colour.
- The supporting text describes the platform as a premium, production-grade AI system that detects leaf pathology in real time and doubles as an IEEE demonstration, portfolio artifact and smart agriculture assistant.
- Two call-to-action buttons, “Analyze Leaf Now” and “View Dashboard”, take the user directly to the diagnosis workflow or to the analytics dashboard.
- The dark background with faint glowing leaf particles reinforces the bioluminescent, neon-green visual identity used consistently across the application.

6.2 Login Page

The Login page authenticates the user before granting access to the diagnosis workstation and dashboard.

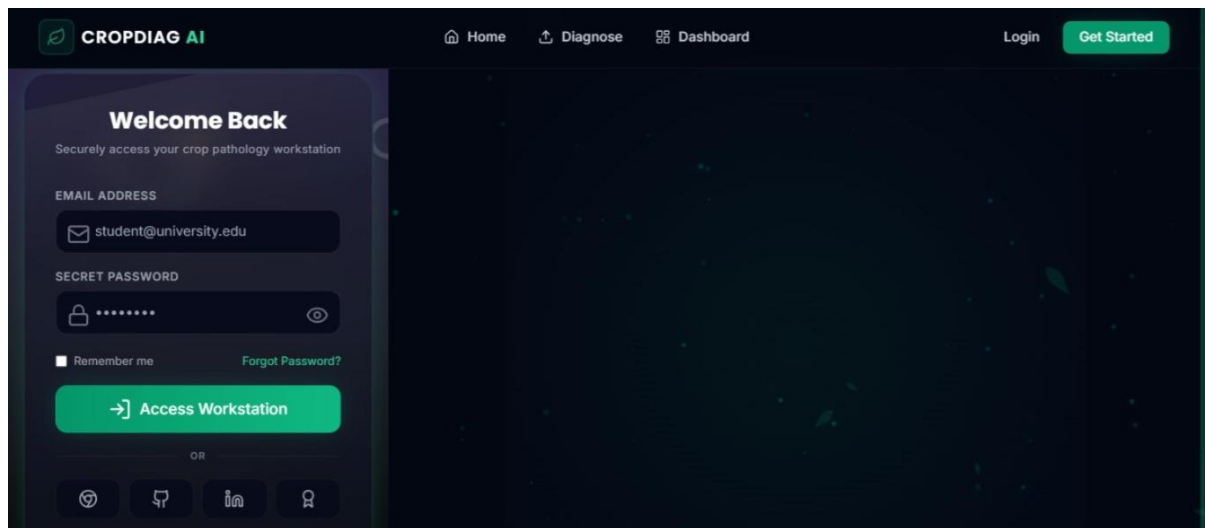


Figure 2: Login Page

- The panel is titled “Welcome Back” with the subtitle “Securely access your crop pathology workstation”, framing the tool as a professional workstation rather than a casual app.
- The Email Address field accepts the registered e-mail (shown with the placeholder student@university.edu) and the Secret Password field masks the entered password, with an eye icon to toggle visibility.
- A “Remember Me” checkbox and a “Forgot Password?” link are provided for returning users.
- The green “Access Workstation” button submits the credentials and signs the user in.
- Below an “OR” divider, four alternate sign-in icons are available for quick authentication through other identity providers.
- The right-hand side of the screen continues the same dark, particle-lit theme as the Home page, keeping the visual identity consistent.

6.3 Disease Prediction Result

After a leaf image is uploaded and analysed, the Diagnostic Result screen presents the CNN model's prediction along with a downloadable report.



Figure 3: Disease Prediction Result

- On the left, the uploaded leaf image is displayed as a thumbnail with a unique Diagnosis ID (#60 in this example) for traceability.
- On the right, a “DIAGNOSTIC RESULT” badge is followed by the predicted class, shown here as “Apple · Healthy”, while a Confidence panel in the top-right corner reports the model's prediction confidence (60% in this example).
- Three summary boxes present Damage Severity, Recovery Window and Status Assessment; for a healthy leaf these read N/A, N/A and “Crop Health Standard Optimal” respectively.
- A “DIAGNOSIS SUMMARY” panel gives a short natural-language explanation of the result — in this case noting that the apple tree is in a highly healthy state with strong shoot growth, bright green leaves and developing fruit.
- Below the summary, editable “Student Name (for report)” and “Project Title” fields allow the result to be personalised before export.
- The green “Download University PDF” button generates a formatted diagnosis report, with additional “Print” and “Share” options alongside it.

6.4 Dashboard & Analytics

The Dashboard provides a real-time, analytics-driven view of overall system performance, sourced from the CNN inference pipeline and the SQLite data registry.

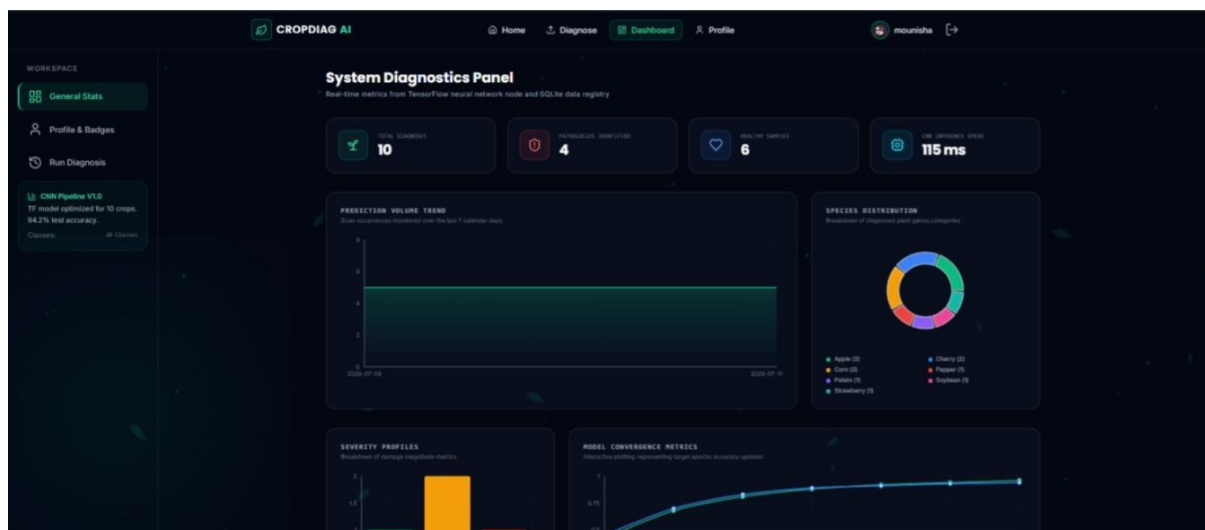


Figure 4: Dashboard & Analytics

- The top navigation bar shows the active user (mounisha) with a profile avatar and logout icon, alongside Home, Diagnose and Dashboard links.
- A left-hand Workspace sidebar offers General Stats, Profile & Badges and Run Diagnosis views, along with a “CNN Pipeline v1.0” info card noting that the TensorFlow model is optimised for 10 crops with 94.2% test accuracy across 20 classes.
- The “System Diagnostics Panel” heading is followed by four key metric cards: Total Diagnoses (10), Pathologies Identified (4), Healthy Samples (6) and CNN Inference Speed (115 ms).
- A “Prediction Volume Trend” area chart plots scan occurrences monitored over the last 7 calendar days, showing usage activity over time.
- A “Species Distribution” donut chart breaks down diagnosed samples by plant genus — Apple, Corn, Potato, Cherry, Pepper, Strawberry and Soybean.
- Further down, “Severity Profiles” and “Model Convergence Metrics” charts summarise damage-magnitude distribution and the model's training accuracy progression across epochs, giving a complete operational snapshot of the AI system.

8. Conclusion

The Automated Crop Disease Diagnosis Using CNN project demonstrates the effective application of deep learning and image processing in agriculture. The system provides an automated workflow in which a crop leaf image is uploaded, preprocessed and analysed by a Convolutional Neural Network to predict the most probable disease class.

The project integrates a modern, bioluminescent-themed user interface, a Flask-based backend and a CNN-based image classification model into a single, cohesive application. The modular architecture — spanning authentication, upload, preprocessing, classification, reporting and dashboard analytics — makes the system easy to understand, maintain and extend. Image preprocessing ensures consistent input to the model, while the CNN learns visual disease patterns such as spots, colour changes and texture variations.

The working screens presented in this report — the Home page, Login page, Disease Prediction Result and Diagnostics Dashboard — together show a complete, end-to-end user journey from authentication through diagnosis to analytics review, giving the project the look and feel of a production-grade smart agriculture platform.

CropDiag AI can support faster preliminary disease identification and can be extended with more crops, larger real-world datasets, multilingual farmer assistance and mobile deployment. The project highlights how Artificial Intelligence can contribute to smart farming and precision agriculture.

PROJECT TITLE: AUTOMATED CROP DISEASE DIAGNOSIS USING CNN

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